

A smoothing-unitary counterexample to a claimed spectral inequality for elliptic pseudodifferential operators

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Status. Candidate counterexample, likely valid, awaiting human review. The claim inspected is Theorem 1.4 of Cardona [1], together with the principal-symbol interpretation of the second open problem on page 18.

1 Source problem and claimed theorem

The source asks in Section 3 (physical page 18) for which manifolds the restriction $\rho \geq 1 - \delta$ can be removed, and conjectures that on a manifold with symmetries, under a suitable positivity condition on the symbol, the spectral inequality should hold throughout $0 \leq \delta < \rho \leq 1$. The same paper’s Theorem 1.4 claims, in the standard range, that every positive elliptic $E \in \Psi_{\rho,\delta}^\nu(M)$ with positive symbol satisfies

$$\|v\|_{L^2(M)} \leq C_1 e^{C_2 \lambda} \|v\|_{L^2(\omega)} \quad \text{for } v \in \text{span}\{\rho_j : \lambda_j \leq \lambda\}.$$

In the proof, the relevant $E(x, \xi)$ is explicitly called the principal symbol. The following construction contradicts this theorem already for $(\rho, \delta) = (1, 0)$ and shows that principal-symbol positivity cannot be the “suitable positivity condition” sought by the open problem.

2 Proof intuition

Ellipticity and the principal symbol only control an operator modulo smoothing terms. Smoothing terms can nevertheless rearrange individual smooth eigenfunctions completely. Start with the standard positive elliptic operator $A = (I + L)^{\nu/2}$ on the circle, whose constant eigenfunction is everywhere visible. A rank-one Householder reflection $U = I + R$, with R smoothing, can send that constant to any prescribed smooth bump f supported away from the sensor. The conjugate $E = UAU^*$ is still positive elliptic and has exactly the same principal symbol as A , but f is now an eigenfunction. The proposed observability estimate then reads $1 \leq 0$.

3 Formal counterexample

Theorem 1. *Let $\nu > 0$. There exist a nonempty open set $\omega \subset \mathbb{S}^1$ and a positive, self-adjoint, invertible elliptic pseudodifferential operator*

$$E \in \Psi_{1,0}^\nu(\mathbb{S}^1)$$

whose principal symbol is the positive symbol $|\xi|^\nu$ for $\xi \neq 0$, but for which the spectral inequality of [1, Theorem 1.4] fails. Indeed, E has a normalized C^∞ eigenfunction f satisfying $f|_\omega = 0$.

$$\| \cdot \|_{H^s(M \times (\alpha, \alpha + \epsilon))} \sim 80,800 \| \cdot \|_{H^s(M_T)} \| \cdot \|_{L^2(\omega)}.$$

In conclusion with this work

- we show that the theory of pseudo-differential operators is an effective microlocal tool for the control theory of parabolic problems, even when we have the lack of Carleman estimates.

The following are some open problems that arise from this investigation.

- To extend the spectral inequality in (2.29) to the case of suitable boundary value problem on compact manifolds with smooth boundary. Just to mention a few, the case of the Laplacian $-\Delta_g$ with the corresponding Dirichlet boundary conditions was derived in the classical Lebeau-Robiano result in [15], and the case of the bi-Laplacian $(-\Delta_g)^2$ endowed with the so called *clamped* boundary conditions was analysed in the recent work of Rousseau and Robiano [24]. Even, in [24] the authors leave open the question of the validity of the spectral inequality for the powers $(-\Delta_g)^k$, $k \geq 3$, when $\partial M \neq \emptyset$.
- The following problem could be of interest in global analysis: to analyse for which kind of manifolds (M, g) the condition $\rho \geq 1 - \delta$ can be removed in Theorem 1.4. Indeed, the inequality $\rho \geq 1 - \delta$ is the required condition in order to have the invariance of the pseudo-differential calculus under changes of coordinates. We propose the following conjecture: in the case where M is a manifold with symmetries, and under a suitable positivity condition on the symbol of the operator $E(x, D)$, for any $0 \leq \delta < \rho \leq 1$, the spectral inequality in Theorem 1.4 is still valid.

4. APPENDIX I: PSEUDO-DIFFERENTIAL OPERATORS AND NULL-CONTROLLABILITY ON HILBERT SPACES

In this section we present the preliminaries about the theory of pseudo-differential

Figure 1: The two open problems in Section 3 of arXiv:2209.10690, rendered from physical page 18. The second bullet is the selected target.

Proof. Give $\mathbb{S}^1$ normalized Haar measure and let $L = -d^2/dx^2 \geq 0$. Put

$$A = (I + L)^{\nu/2}.$$

By the standard functional calculus, A is positive, self-adjoint, invertible, belongs to $\Psi_{1,0}^\nu(\mathbb{S}^1)$, and has principal symbol $|\xi|^\nu$. Its normalized constant function $h \equiv 1$ satisfies $Ah = h$.

Choose a proper open arc ω whose complement has nonempty interior. Choose a real $f \in C^\infty(\mathbb{S}^1)$ with

$$\text{supp } f \subset \mathbb{S}^1 \setminus \omega, \quad \|f\|_2 = 1.$$

Work first on the real Hilbert space. Since $f \neq h$, define

$$w = \frac{h - f}{\|h - f\|_2}, \quad P_w u = \langle u, w \rangle w, \quad U = I - 2P_w.$$

The operator U is a self-adjoint unitary. If $c = \langle h, f \rangle \in \mathbb{R}$ and $d = \|h - f\|_2$, then $d^2 = 2(1 - c)$ and $\langle h, w \rangle = d/2$, $\langle f, w \rangle = -d/2$. Consequently

$$Uh = h - dw = f, \quad Uf = f + dw = h.$$

Extend U complex linearly to complex $L^2(\mathbb{S}^1)$.

Now set

$$E = UAU.$$

Unitary conjugation shows that E is positive, self-adjoint, invertible, and unitarily equivalent to A . It remains to check the microlocal assumptions. The rank-one projection P_w has smooth Schwartz kernel $w(x)\overline{w(y)}$, hence $P_w \in \Psi^{-\infty}$. Expanding gives

$$E - A = -2P_w A - 2AP_w + 4P_w AP_w.$$

Every term on the right has a smooth finite-rank kernel: for example, the kernels of AP_w and $P_w A$ are respectively $(Aw)(x)\overline{w(y)}$ and $w(x)\overline{(Aw)(y)}$. Thus $E - A \in \Psi^{-\infty}$. It follows that $E \in \Psi_{1,0}^\nu$ is elliptic and has the same positive principal symbol as A . Also $U = I + \Psi^{-\infty}$ preserves all Sobolev spaces, so the unbounded-operator domain causes no change of class.

Finally,

$$Ef = UAUf = UAh = Uh = f.$$

Thus f is an eigenfunction of eigenvalue $1 = 1^\nu$ and belongs to the spectral subspace at threshold $\lambda = 1$. The asserted inequality would give

$$1 = \|f\|_{L^2(\mathbb{S}^1)} \leq C_1 e^{C_2} \|f\|_{L^2(\omega)} = 0,$$

a contradiction. □

Remark 1 (Null controllability). *The same invisible eigenmode also obstructs the claimed null controllability: the observation/control multiplier 1_ω annihilates f , so the adjoint observability inequality fails on the one-dimensional invariant subspace spanned by f .*

4 Verification and scope

The construction is exact and uses no computation. The decisive checks are: (i) a smooth rank-one kernel is smoothing; (ii) smoothing perturbations preserve the principal symbol and ellipticity; (iii) unitary conjugation preserves positivity and spectrum; and (iv) the Householder identities swap h and f . All four are displayed explicitly in the proof.

This result does not solve the source’s first, boundary/polyharmonic open problem. It also does not refute the deliberately unspecified existential phrase “under a suitable positivity condition.” Rather, it proves sharply that the hypotheses of source Theorem 1.4—positive operator and positive principal symbol—are insufficient. The later compact-Lie-group theorem of Cardona, Delgado and Ruzhansky [2] assumes nonnegativity of the full global matrix-valued symbol. Our construction is not claimed to satisfy that stronger, quantization-specific condition.

5 Attempt history and novelty check

An initial route observed that Navier powers of the Dirichlet Laplacian inherit the Laplacian spectral inequality, but that proper subcase does not reach the clamped boundary conditions in the first open problem. The smoothing-unitary route upgrades the investigation to the complete counterexample above.

The bounded search on 11 August 2026 used the exact source title and arXiv id with the terms *correction*, *erratum*, and *counterexample*, and searched close variants involving spectral inequalities, elliptic pseudodifferential operators, polyharmonic boundary conditions, and Lebeau–Robbiano estimates. No correction or matching counterexample was found. The close works arXiv:2209.12092 [2] and arXiv:2309.02181 were inspected. Novelty confidence is moderate because finite-rank smoothing conjugation is standard folklore even though this exact application was not located.

6 Human review recommendation

A microlocal-analysis reviewer should focus on the source's interpretation of $E(x, \xi) > 0$. Its proof explicitly identifies this as the principal symbol; under that reading Theorem 1 is complete. If a stronger complete-symbol condition was intended, it must be stated separately and the scope of the source theorem changes.

References

- [1] D. Cardona, *Spectral inequalities for elliptic pseudo-differential operators on closed manifolds*, arXiv:2209.10690 (2022).
- [2] D. Cardona, J. Delgado, and M. Ruzhansky, *Estimates for sums of eigenfunctions of elliptic pseudo-differential operators on compact Lie groups*, arXiv:2209.12092 (2022; revised 2022).
- [3] L. Hörmander, *The Analysis of Linear Partial Differential Operators III*, Springer, 1985.